
3.4.1 The dynamic equilibrium of populations is affected by a number of factors.

Populations and ecosystems

A population is all the organisms of one species in a habitat. Populations of different species form a community.

Within a habitat a species occupies a niche governed by adaptation to both biotic and abiotic conditions.

Investigating populations

A critical appreciation of some of the ways in which the numbers and distribution of organisms may be investigated.

Random sampling with quadrats and counting along transects to obtain quantitative data.

The use of percentage cover and frequency as measures of abundance.

The use of mark–release–recapture for more mobile species.

In the context of investigating populations, **candidates should**

- carry out experimental and investigative activities, including appropriate risk management
 - consider ethical issues arising when carrying out fieldwork, particularly those relating to the organisms involved and their environment
 - analyse and interpret data relating to the distribution of organisms, recognising correlations and causal relationships
 - appreciate the tentative nature of conclusions that may be drawn from such data.
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Variation in population size

Population size may vary as a result of

- the effect of abiotic factors
 - interactions between organisms: interspecific and intraspecific competition and predation.
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Human populations

Population size and structure, population growth rate, age population pyramids, survival rates and life expectancy.

Candidates should be able to

- interpret growth curves, survival curves and age pyramids
 - calculate population growth rates from data on birth rate and death rate.
 - relate changes in the size and structure of human populations to different stages in demographic transition.
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